

```
root@NewZone:~# reboot
Aug 8 14:15:37 NewZone reboot: initiated by root on /dev/console

[NOTICE: Zone rebooting]

SunOS Release 5.11 Version oi_151 64-bit
Copyright (c) 1983, 2010, Oracle and/or its affiliates. All rights reserved.
Hostname: NewZone

NewZone console login: root
Password:
Aug 8 14:16:18 NewZone login: ROOT LOGIN /dev/console
Last login: Mon Aug 8 14:09:54 on console
OpenIndiana (powered by illumos) SunOS 5.11 oi_151 July 2011
root@NewZone:~# date
Mon Aug 8 14:16:22 CEST 2011
```

I propose to build a QEMU package, so it can be installed later inside *NewZone*. Why is this so important? It could sometimes be essential to have a standalone package, if it's new and not in the OI repository yet. Or, in the opposite case—for your own corporate program which you don't wish to redistribute. As an example, let's analyse how to create a package from the set of programs in */opt/qemu-0.14*. The first step is to prepare two files, *prototype* and *pkginfo*.

```
# echo 'i pkginfo' > prototype
# pkgproto /opt/qemu-0.14/ >> prototype
```

In *prototype*, you should specify those files and directories that the future package will contain. The *pkginfo* file will contain all information about this package:

```
root@openindiana:~/pkg# cat pkginfo
BASEDIR=/opt/qemu-0.14
TZ=PST
PATH=/sbin:/usr/sbin:/usr/bin:/usr/sadm/install/bin
PKG=qemu
ARCH=x86
NAME=qemu
VERSION=0.14
CATEGORY=application
DESC=QEMU virtualization
VENDOR=QEMU.ORG
EMAIL=qemu@qemu.org
PKGSAP=/var/sadm/pkg/qemu/save
```

Begin building the package itself:

```
root@openindiana:~/pkg# pkgmk -b /opt/qemu-0.14 -r /
## Building pkgmap from package prototype file.
## Processing pkginfo file.
```

```
WARNING: missing directory entry for <opt>
WARNING: parameter <PSTAMP> set to "openindiana20110814071557"
WARNING: parameter <CLASSES> set to "none"
## Attempting to volumize 84 entries in pkgmap.
part 1 -- 123944 blocks, 428 entries
## Packaging one part.
/var/spool/pkg/qemu/pkgmap
/var/spool/pkg/qemu/pkginfo
/var/spool/pkg/qemu/root/opt/qemu-0.14/bin/qemu
[...]
/var/spool/pkg/qemu/root/opt/qemu-0.14/share/qemu/vgabios-stdvga.
bin
/var/spool/pkg/qemu/root/opt/qemu-0.14/share/qemu/vgabios-vmware.
bin
/var/spool/pkg/qemu/root/opt/qemu-0.14/share/qemu/vgabios.bin
## Validating control scripts.
## Packaging complete.
```

The package is ready, and is stored in */var/spool/pkg*. The last step is to convert it to a special *pkg* format:

```
root@openindiana:~/pkg# pkgtrans -s /var/spool/pkg/ qemu.pkg
```

The following packages are available:

```
1 qemu      qemu
           (x86) 0.14
```

```
Select package(s) you wish to process (or 'all' to process
all packages). (default: all) [?,??,q]: 1
Transferring <qemu> package instance
root@openindiana:~/pkg# ls -l /var/spool/pkg/
total 58668
drwxr-xr-x 3 root root      5 2011-08-14 08:15 qemu
-rw-r--r-- 1 root root 59996160 2011-08-14 08:19 qemu.pkg
```

Now, transfer *qemu.pkg* to the isolated area, and install it in *NewZone*:

```
root@NewZone:/export# pkgadd -d qemu.pkg
```

The following packages are available:

```
1 qemu      qemu
           (x86) 0.14
```

```
Select package(s) you wish to process (or 'all' to process
all packages). (default: all) [?,??,q]: 1
```

```
Processing package instance <qemu> from </export/qemu.pkg>
```

```
qemu(x86) 0.14
QEMU.ORG
## Processing package information.
## Processing system information.
## Verifying disk space requirements.
```